Hypothesis Testing — Cheat Sheet

Last-minute revision guide · Based on: Mahbub Latif, PhD — Hypothesis Testing (Feb 2026)

1. Core Concepts

Statistical hypothesis — a claim about a population parameter, not known a priori to be true.

- **H_0 (null):** hypothesis being tested. Simple ($\theta=\theta_0$) or Composite ($\theta>\theta_0$)
- **H_1 (alternative):** accepted if H_0 rejected
- **Critical region C:** reject H_0 if sample $\in C$

	H_0 True	H_0 False
Reject H_0	Type I error (α) false positive	Correct ✓
Accept H_0	Correct ✓	Type II error (β) false negative

$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$ — fixed in advance (0.10, 0.05, 0.005)
$\beta(\mu) = P(\text{accept } H_0 \mid H_0 \text{ false}) \rightarrow$ plot vs μ = OC curve
Power $= 1 - \beta(\mu)$

p-value = smallest α at which H_0 can still be rejected.
Reject H_0 if p-value ≤ α.

2. Mean Test — σ^2 KNOWN (Z-test)

Population: $N(\mu, \sigma^2)$, σ known. Sample: $X_1,\ldots,X_n$

$$Z = (\bar{X} - \mu_0) / (\sigma/\sqrt{n}) \sim N(0,1) \text{ under } H_0$$

H_0	H_1	Reject if	p-value
$\mu=\mu_0$	$\mu\neq\mu_0$	$\lvert Z\rvert > z_{\alpha/2}$	$2P(Z\geq\lvert z\rvert)$
$\mu=\mu_0$	$\mu>\mu_0$	$Z > z_\alpha$	$P(Z\geq z)$
$\mu=\mu_0$	$\mu<\mu_0$	$Z < -z_\alpha$	$P(Z\leq z)$

$z_{0.10}=1.28$ | $z_{0.05}=1.645$ | $z_{0.025}=1.96$ | $z_{0.01}=2.33$ | $z_{0.005}=2.576$

OC (2-sided): $\beta(\mu_1) = \Phi(\,(\mu_0-\mu_1)/(\sigma/\sqrt{n}) + z_{\alpha/2}\,) - \Phi(\,(\mu_0-\mu_1)/(\sigma/\sqrt{n}) - z_{\alpha/2}\,)$
OC (1-sided, H_1:$\mu>\mu_0$): $\beta(\mu_1) = \Phi(\,z_\alpha + (\mu_0-\mu_1)/(\sigma/\sqrt{n})\,)$

CI link: reject H_0:$\mu=\mu_0$ at level $\alpha \iff \mu_0$ lies outside $\bar{X} \pm z_{\alpha/2}\cdot\sigma/\sqrt{n}$

3. Mean Test — σ^2 UNKNOWN (t-test)

Estimate: $S^2 = \Sigma(X_i-\bar{X})^2 / (n-1)$

$$T = (\bar{X} - \mu_0) / (S/\sqrt{n}) \sim t_{n-1} \text{ under } H_0$$

H_0	H_1	Reject if	p-value
$\mu=\mu_0$	$\mu\neq\mu_0$	$\lvert T\rvert > t_{n-1,\alpha/2}$	$2P(T\geq\lvert t\rvert)$
$\mu=\mu_0$	$\mu>\mu_0$	$T > t_{n-1,\alpha}$	$P(T\geq t)$
$\mu=\mu_0$	$\mu<\mu_0$	$T < -t_{n-1,\alpha}$	$P(T\leq t)$

Rule of thumb: σ unknown → **always t-test**, df = n−1

4. Two-Sample Mean Tests

H_0: $\mu_x=\mu_y$ vs H_1: $\mu_x\neq\mu_y$ | $\bar{X}$ (n obs), $\bar{Y}$ (m obs)

Case I — σ_x^2, σ_y^2 known:

$$Z = (\bar{X}-\bar{Y}) / \sqrt{(\sigma_x^2/n + \sigma_y^2/m)} \sim N(0,1)$$

Case II — $\sigma_x^2=\sigma_y^2=\sigma^2$ unknown (pooled t):

$$S_p^2 = [(n-1)S_x^2+(m-1)S_y^2] / (n+m-2)$$
$$T = (\bar{X}-\bar{Y}) / [S_p\sqrt{(1/n+1/m)}] \sim t_{n+m-2}$$

Case III — unknown & unequal, large n,m:

$$Z = (\bar{X}-\bar{Y}) / \sqrt{(S_x^2/n + S_y^2/m)} \sim N(0,1) \text{ approx.}$$

Paired t-test (dependent samples, e.g. before/after):

$$W_i = X_i - Y_i,\ H_0: \mu_w = 0$$
$$Z = \bar{W} / (S_w/\sqrt{n}) \sim N(0,1), \text{ reject if } \lvert Z\rvert > z_{\alpha/2}$$

Use **paired** test when it's the *same subjects* measured twice — not independent groups.

5. Variance Tests

One population: H_0: $\sigma^2=\sigma_0^2$ vs H_1: $\sigma^2\neq\sigma_0^2$

$$\chi^2 = (n-1)S^2 / \sigma_0^2 \sim \chi^2_{n-1} \text{ under } H_0$$

Reject if $\chi^2 > \chi^2_{n-1,\alpha/2}$ **or** $\chi^2 \leq \chi^2_{n-1,1-\alpha/2}$

χ^2 is **NOT symmetric** — the two tail cutoffs differ; always look up both separately.

Two populations (F-test): H_0: $\sigma_x^2=\sigma_y^2$

$$F = S_x^2/S_\gamma^2 \sim F_{n-1,m-1} \text{ under } H_0$$

Reject if $F < F_{n-1,m-1,1-\alpha/2}$ **or** $F > F_{n-1,m-1,\alpha/2}$

$p = 2\times\min\{P(F<f),\ 1-P(F<f)\}$

6. Bernoulli / Proportion Tests

One proportion: $X \sim B(n,p)$, $H_0: p \le p_0$ vs $H_1: p > p_0$

Exact test (small n or $np_0 \le 5$):

$$\text{p-value} = P(X \ge x \mid p=p_0) = \sum_{i=x}^{n} C(n,i) p_0^i (1-p_0)^{n-i}$$

Approx test (valid if $np_0 > 5$ AND $n(1-p_0) > 5$):

$$Z = (X - np_0) / \sqrt{np_0(1-p_0)} \sim N(0,1) \text{ approx.}$$

Two-sided: reject if $|Z| > z_{\alpha/2}$; one-sided reject if $Z > z_\alpha$

Two proportions: $H_0: p_1 = p_2$, $X \sim B(n,p_1)$, $Y \sim B(m,p_2)$

Exact (Fisher-Irwin): condition on $X+Y=k$ → hypergeometric

$$P(X=i \mid X+Y=k) = C(n,i)C(m,k-i) / C(n+m,k)$$

Approx test:

$$\hat{p} = (X+Y)/(n+m)$$
$$Z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n+1/m)]} \sim N(0,1)$$

7. Quick Decision Map

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Testing a MEAN?
 ├─ σ known?                              → Z-test  (σ/√n)
 ├─ σ unknown?                            → t-test  (S/√n, df = n−1)
 ├─ Two samples, σ's known?               → Z-test (two-sample)
 ├─ Two samples, σ's unknown, equal?      → Pooled t-test (df = n+m−2)
 ├─ Two samples, σ's unknown, unequal, large n,m?  → Approx Z-test
 └─ Paired / same subjects twice?         → Paired t-test on differences W

Testing a VARIANCE?
 ├─ One population   → Chi-square test (df = n−1)
 └─ Two populations  → F-test (df = n−1, m−1)

Testing a PROPORTION?
 ├─ One sample, n small / extreme p₀        → Exact Binomial test
 ├─ One sample, np₀>5 and n(1−p₀)>5         → Approx Z-test
 ├─ Two samples, small                      → Fisher-Irwin (hypergeometric) exact test
 └─ Two samples, large                      → Approx Z-test (pooled p̂)
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8. Key Reminders for the Exam

1. **Always state H_0 and H_1 first**, then identify the correct test statistic.
2. **Two-sided test** → split α into two tails (α/2 each); compare |test stat| to critical value.
3. **One-sided test** → all of α in one tail; direction of H_1 tells you which tail.
4. **p-value ≤ α ⟺ reject H_0.** Smaller p-value = stronger evidence against H_0.
5. Confidence interval and two-sided hypothesis test are equivalent: reject H_0 iff μ_0 is outside the CI.
6. Chi-square / F critical values are **not symmetric** — always look up both tail values separately.
7. Paired design (same subject, before/after) ≠ independent two-sample design — mixing these up is a common exam mistake.
8. For proportions, **always check $np_0 > 5$ and $n(1-p_0) > 5$** before using the normal approximation; otherwise use the exact binomial test.